Supplementary Information

Charge change predicts the measured effect of a pH-responsive protein design ladder in acid-sensing ion channel 1a

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Table S1. Measured activation shift by structural domain, using the domain assignment of the source study.

Structural domain	Substitutions	Mean |shift| in activation pH50 (pH units)
Finger and thumb	8	0.512
Palm	49	0.303
Acidic pocket	45	0.281
β-ball	4	0.112
Knuckle	8	0.058
Transmembrane	3	0.057

[1]Substitutions scored in that domain out of the 117 in the primary cohort.
[2]Mean of the absolute shift in activation pH50 against the cohort wild type. The domains differ at Kruskal-Wallis $p = 0.0001$. The two extreme domains carry the fewest substitutions, so the ordering of the extremes is less secure than the difference between domains.

Table S2. Acidic-pocket carboxylate pair geometry across four biological assemblies.

Entry	Species	Structure pH	Carboxylate pair	Minimum intra-subunit O to O distance (A)	Closest inter-subunit equivalent (A)	Margin (A)
9E4A	human	8.5	D237-D351	10.69	49.54	38.84
9E4A	human	8.5	E238-D347	7.55	51.41	43.86
9E4A	human	8.5	E219-D409	4.61	27.79	23.18
9E4A	human	8.5	E79-E418	4.87	11.13	6.25
9E4F	human	5.7	D237-D351	3.2	43.74	40.53
9E4F	human	5.7	E238-D347	3.1	46.8	43.7
9E4F	human	5.7	E219-D409	4.78	27.07	22.29
9E4F	human	5.7	E79-E418	3.66	5.19	1.54
6VTL	chicken	8.0	D238-D350	10.46	51.52	41.07
6VTL	chicken	8.0	E239-D346	9.78	47.01	37.23
6VTL	chicken	8.0	E220-D408	3.39	28.15	24.76
6VTL	chicken	8.0	E80-E417	3.95	12.83	8.88
2QTS	chicken	6.5	D238-D350	2.46	44.47	42.01
2QTS	chicken	6.5	E239-D346	2.59	48.44	45.85
2QTS	chicken	6.5	E220-D408	2.54	26.19	23.64
2QTS	chicken	6.5	E80-E417	3.24	5.25	2.01

[1]The pH at which the structure was determined, as deposited.
[2]Minimum carboxyl oxygen to carboxyl oxygen distance between the two members of the pair within a single subunit.
[3]The same distance measured to the nearest copy of the partner residue in a different subunit.
[4]Inter-subunit minus intra-subunit. Every value is positive, so both members of every pair come from one subunit in every assembly; the smallest margin is 1.54 angstrom. The 6VTL assembly is truncated as deposited and 42 of its 60 aspartate and glutamate residues carry both carboxyl oxygens, so distances for that entry are upper bounds.

Table S3. Predicted pKa of the nine acidic-pocket carboxylates in two structural states of the same human construct.

Position	Residue	Predicted pKa, 9E4F (pH 5.7)	Buried (%), 9E4F	Predicted pKa, 9E4A (pH 8.5)	Buried (%), 9E4A	Change between states	M co pK
79	E79	6.94	100	4.92	88	2.02	4.5
219	E219	5.9	99	6.98	93	-1.08	4.5
237	D237	8.06	87	3.96	0	4.1	3.8
238	E238	11.82	100	6.52	65	5.3	4.5
242	E242	7.74	100	5.58	100	2.16	4.5
347	D347	5.2	93	5.23	50	-0.03	3.8
351	D351	3.95	83	4.29	23	-0.34	3.8
409	D409	9.19	100	9.32	100	-0.13	3.8
418	E418	8.8	100	5.65	79	3.15	4.5

[1]PROPKA3 on the assembled trimer of 9E4F, determined at pH 5.7.
[2]PROPKA3 on the assembled trimer of 9E4A, determined at pH 8.5.
[3]Low-pH value minus resting value. 5 of the 9 carboxylates move by more than two units between the two states and the largest move is 5.30 units, so each predicted value describes the state it was computed on. Predicted perturbation does not rank the measured activation sensitivity of the 23 carboxylate positions carrying both (Spearman rho +0.094, p = 0.669).

Table S4. Residue numbering between the chicken structures and human ASIC1a, by global pairwise alignment.

Human ASIC1a	Chicken structures	Offset	Role in this work
H73	H74	+1	measured position
D78	D79	+1	measured position
K211	K212	+1	measured position
E219	E220	+1	acidic pocket pair 3
D237	D238	+1	acidic pocket pair 1
E238	E239	+1	acidic pocket pair 2
E242	E243	+1	measured position
A243	A244	+1	measured position
D347	D346	-1	acidic pocket pair 2
D351	D350	-1	acidic pocket pair 1
E375	E374	-1	measured position
D409	D408	-1	acidic pocket pair 3
E413	E412	-1	measured position
E418	E417	-1	acidic pocket pair 4
G433	G432	-1	measured position
D434	D433	-1	measured position

[1]Chicken position minus human position. The offset takes the values -1 and +1, changing sign inside the extracellular domain, so no single global offset maps the pocket correctly. Every row was checked to carry the same residue identity in both species; a mismatch is a fatal condition in the script that builds this table.

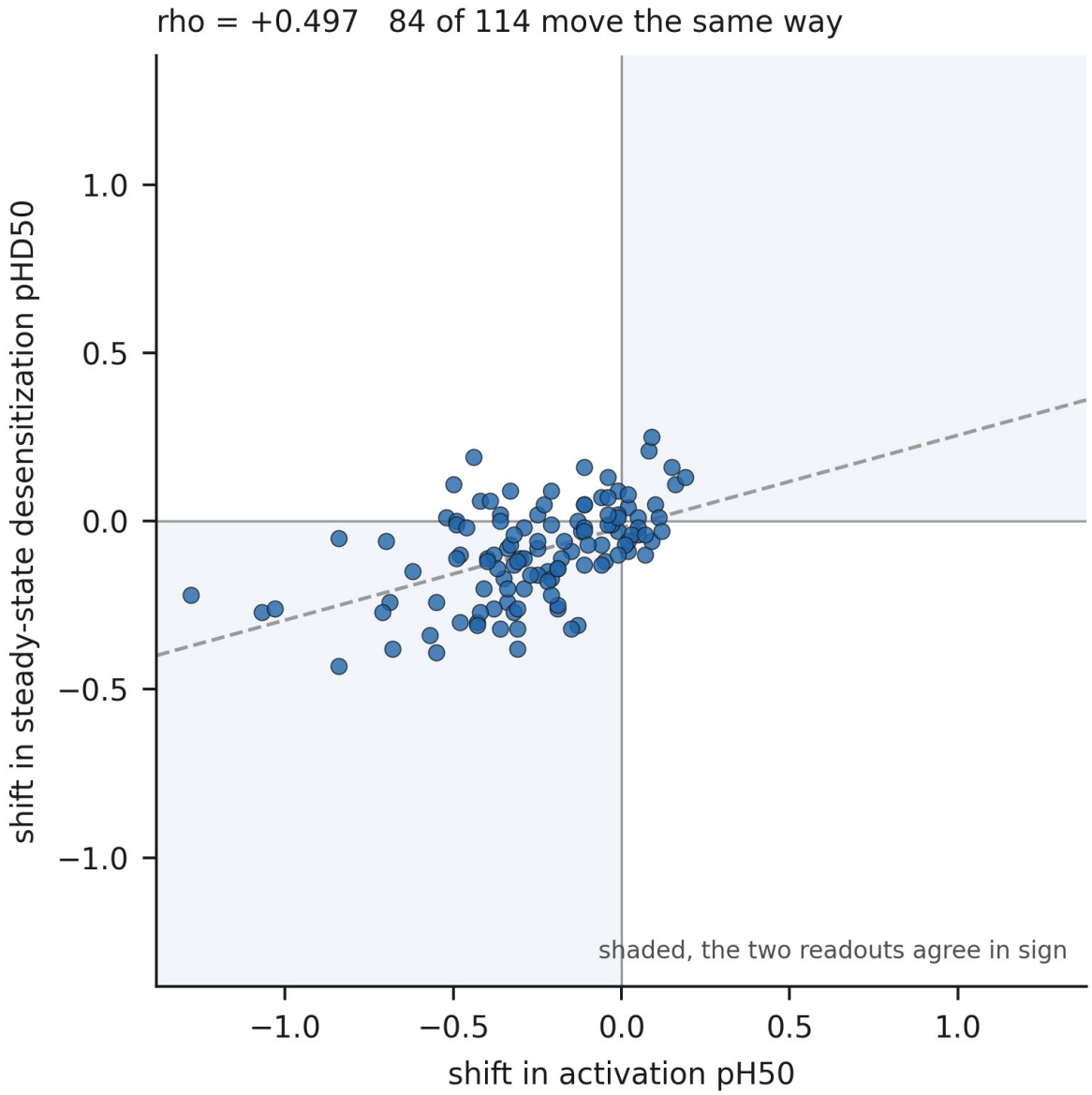


Figure S1. The two readouts are coupled.

[1]Signed shift in steady-state desensitization pHD50 against signed shift in activation pH50, for the 117 substitutions carrying both. Spearman rho +0.497, permutation $p < 0.0001$; 84 of 114 substitutions move the two midpoints in the same direction. The fixed-effects model was fitted to each readout separately, and the agreement of the two null results is weaker evidence than two independent nulls would be.

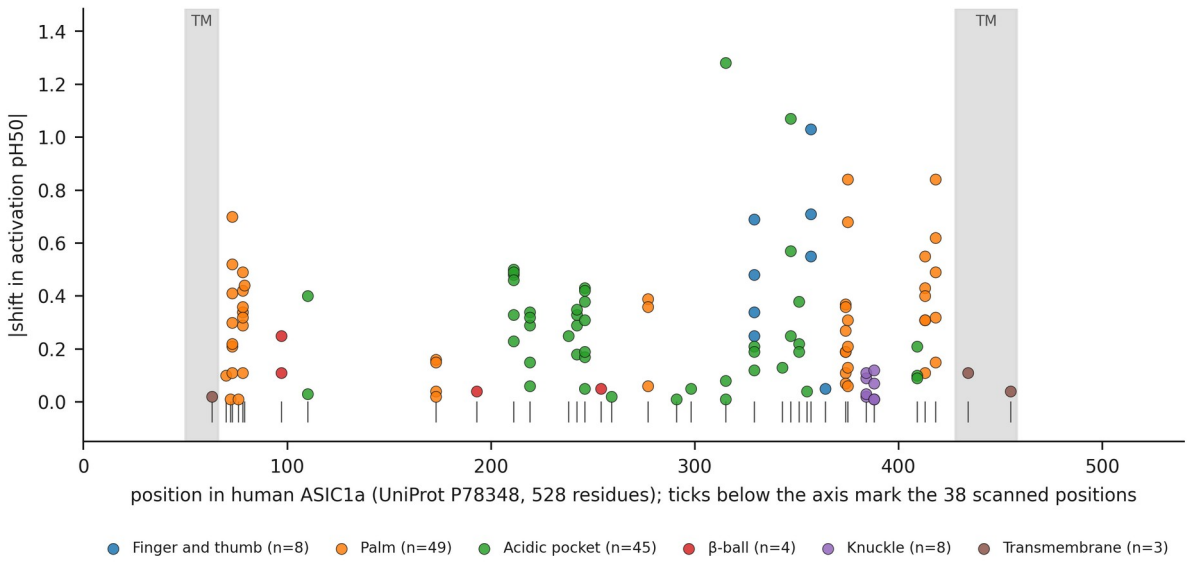


Figure S2. Measured shift along the sequence.

[1]Absolute shift in activation pH50 for every substitution, placed at its position in human ASIC1a and coloured by the structural domain the source assigns. The transmembrane segments are shaded. The scanned positions are titratable residues chosen as candidate proton sensors and are not a uniform sample of the sequence.

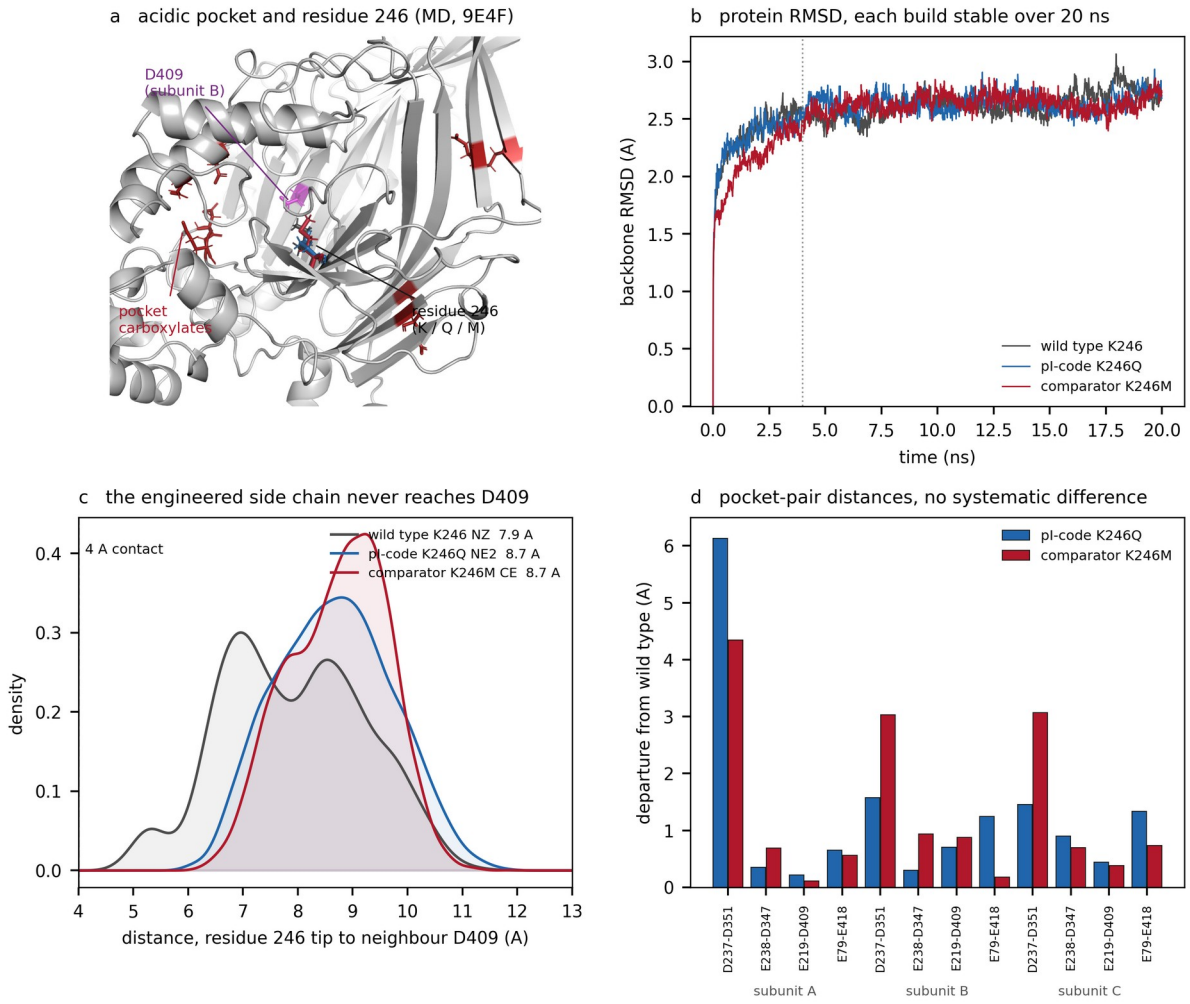


Figure S3. Molecular dynamics control at position 246.

[1]Three 20 ns fixed-protonation trajectories of the extracellular domain trimer of 9E4F: wild type (K246), the pI-code ladder substitution (K246Q) and an off-ladder comparator (K246M). Distances are minimum carboxyl oxygen to carboxyl oxygen for four acidic-pocket pairs in each of the three subunits, twelve measurements per build, over the final 80 percent of each trajectory. Mean backbone RMSD is 2.62, 2.64 and 2.63 angstrom. The mean departure from wild type over the twelve pairs is 1.275 angstrom for K246Q and 1.301 for K246M, the ladder substitution lying closer in 5 of 12 pairs (sign test p 0.77, Wilcoxon signed-rank p 1.00); the 0.026 angstrom difference between the two is carried by the D237-D351 pairs, which never come within 4 angstrom in any build and so are not contacts here. Averaged over the three interfaces the tip atom of residue 246 sits 7.92, 8.66 and 8.73 angstrom from D409 of the neighbouring subunit, the closest approach in any frame being 4.66, 5.99 and 6.18 angstrom, so no build forms a contact there and the wild-type lysine is the one that comes nearest. With one trajectory per build and the three subunits as the only replication, this tests whether either substitution remodels the pocket, not what either does to pH50.